

→ FIRST and FOLLOW :

FIRST and FOLLOW are very important concept in Top-Down parsing, especially for constructing an LL(1) parsing table.

→ FIRST :

The FIRST(X) of a grammar symbol X is the set of the terminals that can appear as the first symbol of a string derived from X.

In simple words :

FIRST tells us what symbol can appear at the beginning of something generated by X.

Eg (1) Consider:

$$A \rightarrow aB$$

The first symbol produced by A is a.

$$\therefore \text{FIRST}(A) = \{a\}$$

Eg (2) $A \rightarrow bC$

$$\text{FIRST}(A) = \{b\}$$

Eg (3)

$$A \rightarrow B$$

$$B \rightarrow \bullet c$$

Since, $A \rightarrow B \rightarrow c$

the first terminal generated by A is c.

$$\text{FIRST}(A) = \{c\}$$

→ FIRST of a terminal :

if $[a]$ is a terminal :

$$\text{FIRST}(a) = \{a\}$$

Eg :

$$\text{FIRST}(id) = \{id\}$$

$$\text{FIRST}(+) = \{+\}$$

$$\text{FIRST}('(') = \{(\}$$

($\rightarrow$ open bracket)

A terminal is always the first thing it can produce.

→ FIRST of ϵ :

If A can produce ϵ .
 $A \rightarrow \epsilon$

$$\text{FIRST}(A) = \{\epsilon\}$$

→ Rules of finding FIRST : ✓

These rules are important for find FIRST :

Rule 1: If X is a terminal.

$$\text{FIRST}(X) = \{X\}$$

Rule 2: If $X \rightarrow \epsilon$

$$\text{FIRST}(X) = \{\epsilon\}$$

Rule 3: If $X \rightarrow YZ$

First find $\text{FIRST}(Y)$.

If Y cannot produce ϵ :

$$\text{FIRST}(X) = \text{FIRST}(Y)$$

- If Y can produce ϵ , then also consider Z .

Rule 4: If multiple products exist:

$$A \rightarrow \alpha | \beta | \gamma$$

We calculate FIRST of each alternative and take their Union.

$$\text{FIRST}(A) = \text{FIRST}(\alpha) \cup \text{FIRST}(\beta) \cup \text{FIRST}(\gamma)$$

Eg; ①

$$S \rightarrow AB$$

$$A \rightarrow a | \epsilon$$

$$B \rightarrow b$$

find FIRST (S).

→ We have

$$S \rightarrow AB$$

First symbol is A.

$$\text{FIRST}(A) = \{a, \epsilon\}$$

Since A can produce ϵ , we also look at B.

$$S \rightarrow \overset{\epsilon}{A} B$$

$$\text{FIRST}(B) = \{b\}$$

$$\therefore \text{FIRST}(S) = \{a, b\}$$

Eg(2)

$$S \rightarrow AB$$

$$A \rightarrow a | \epsilon$$

$$B \rightarrow b | \epsilon$$

$$S \rightarrow AB$$

first symbol is A.

$$\text{First}(A) = \{a, \epsilon\}$$

Since A can produce ϵ ; we also look at B

$$S \xrightarrow{\epsilon} AB$$

$$\text{FIRST}(B) = \{b, \epsilon\}$$

Since B can produce ϵ , we also look at empty string (ϵ).

$$S \xrightarrow{\epsilon \epsilon} \cancel{AB}$$

here S produce ϵ .

$$\text{FIRST}(S) = \{a, b, \epsilon\}$$

eg ③ $S \rightarrow ABC$

$$A \rightarrow a | \epsilon$$

$$B \rightarrow b | \epsilon$$

$$C \rightarrow c$$

$S \rightarrow ABC$

first symbol is A

$$\text{FIRST}(A) = \{a, \epsilon\}$$

Since, A can produce ϵ , we can also look at B.

$$\text{FIRST}(B) = \{b, \epsilon\}$$

Since B can produce ϵ , we can look at C.

$$\text{FIRST}(C) = \{c\}$$

$$\therefore \text{FIRST}(S) = \{a, b, c\}$$

Eg ④

$$S \rightarrow ABC \mid d \mid e$$

$$A \rightarrow a \mid \epsilon$$

$$B \rightarrow b \mid gh$$

$$C \rightarrow c \mid \epsilon$$

→

$$S \rightarrow ABC \mid d \mid e$$

FIRST symbol are A, d, e

here, d and e are terminal

$\therefore$

$$\text{FIRST}(A) = \{a, \epsilon\}$$

Since A can produce ϵ , we can look at B .

$$\text{FIRST}(B) = \{b\}$$

Hence,

$$\text{FIRST}(S) = \{a, b, d, e\}$$

Eg (5) $S \rightarrow AB$

$$A \rightarrow CD$$

$$C \rightarrow c | \epsilon$$

$$D \rightarrow d | \epsilon$$

$$B \rightarrow b | \epsilon$$

$\Rightarrow$

$$S \rightarrow AB$$

$$\text{FIRST}(A) = \text{FIRST}(C)$$

$$\text{FIRST}(C) = \{c, \epsilon\}$$

$$\text{FIRST}(D) = \{d, \epsilon\}$$

$$\text{FIRST}(A) = \{c, d, \epsilon\}$$

$$\text{FIRST}(S) = \text{FIRST}(A)$$

here A can produce ϵ , we can look at B

$$\text{FIRST}(B) = \{b, \epsilon\}$$

$$\text{FIRST}(S) = \{b, c, d, \epsilon\}$$

Q. Consider the grammar

$$E \rightarrow TE_1, E_1 \rightarrow +TE_1 \mid \epsilon$$

$$T \rightarrow FT_1, T_1 \rightarrow *FT_1 \mid \epsilon$$

$$F \rightarrow (E) \mid id$$

⇒ Find the FIRST set of all the variables.

All the variables:

$$E, E_1, T, T_1, F$$

let's start:

$$E \rightarrow TE_1$$

$$\text{FIRST}(E) = \text{FIRST}(T)$$

$$\text{FIRST}(T) = \text{FIRST}(F)$$

$$\text{FIRST}(F) = \{ (, id \}$$

$$\text{FIRST}(T) = \{ (, id \}$$

$$\text{FIRST}(E) = \{ \epsilon, c, id \}$$

$$\text{FIRST}(E_1) = \{ +, \epsilon \}$$

$$\text{FIRST}(T_1) = \{ *, \epsilon \}$$

Q. Find the FIRST from the following production.

$$S \rightarrow aBDh, B \rightarrow cC, C \rightarrow bC \mid \epsilon, D \rightarrow EF, E \rightarrow g \mid \epsilon, F \rightarrow f \mid \epsilon.$$

$\Rightarrow$

~~FIRST(S)~~ All the variables / non-terminal

$S, B, C, D, E, F.$

$$\text{FIRST}(S) = \{ a \}$$

$$\text{FIRST}(B) = \{ c \}$$

$$\text{FIRST}(C) = \{ b, \epsilon \}$$

$$\text{FIRST}(D) = \text{FIRST}(E)$$

$$\text{FIRST}(E) = \{ g, \epsilon \}$$

Since, E can produce ϵ , we can look at F

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 $\boxed{\text{FIRST}(F) = \{f, \epsilon\}}$

here, F can produce ϵ ,

$\therefore$

$\boxed{\text{FIRST}(D) = \{g, f, \epsilon\}}$

→ FOLLOW:~

The FOLLOW(A) of a non-terminal A is the set of terminals that can appear immediately after A in some valid derivation.

In simple words;

FOLLOW tells us what can come immediately after a non-terminal.

→ What is \$?

\$ represent the end of input.

for eg;

id + id \$

The \$ tells the parser:

The input has ended.

Important Rule:

The start symbol always contains \$ in its FOLLOW set.

if:

$S = \text{Start Symbol}$

then

$\$ \in \text{FOLLOW}(S)$

Rules for finding FOLLOW:

Rule 1: Start Symbol

If S is the Start symbol.

$\$ \in \text{FOLLOW}(S)$

Rule 2: $A \rightarrow \text{~~ABC~~ DBC}$

B is immediate after D

~~If non-terminal B is followed by P :~~

~~ABC~~

then: $\text{FOLLOW}(D) = \text{FIRST}(B)$

~~$\text{FIRST}(B) \subseteq \text{FOLLOW}(D)$~~

[$\epsilon \rightarrow$ never exist in FOLLOW]

Rule 3: $A \rightarrow \alpha B$

If B is at the end:

$$A \rightarrow \alpha B$$

$$\text{FOLLOW}(A) \subseteq \text{FOLLOW}(B) \quad \{ \text{FOLLOW}(B) = \text{FOLLOW}(A) \}$$

Rule 4: $A \rightarrow \alpha B \beta$ can produce ϵ

$$A \rightarrow \alpha B \beta$$

$$\epsilon \in \text{FIRST}(B)$$

Then:

$$\text{FOLLOW}(A) \subseteq \text{FOLLOW}(B)$$

$$\text{Eg; } ① S \rightarrow \alpha A b$$

Here, b comes 'immediate after' A.

$$\therefore \text{FOLLOW}(A) = \{ b \}$$

$$\text{FOLLOW}(S) = \{ \$ \} \text{ because } S \text{ is start symbol}$$

Eg ②

$$S \rightarrow aA$$

We know:

$$\text{FOLLOW}(S) = \{ \$ \}$$

~~FOLOW~~ Here A is end

There is nothing written after A.

$$\text{FOLLOW}(S) \subseteq \text{FOLLOW}(A) \quad \{ \text{FOLLOW}(A) = \text{FOLLOW}(S) \}$$

$$\therefore \text{FOLLOW}(A) = \{ \$ \}$$

Eg ③

$$S \rightarrow Ab$$

b comes immediate after a

$$\text{FOLLOW}(A) = \{ b \}$$

$$\text{FOLLOW}(S) = \{ \$ \}$$

Eg ④ $S \rightarrow AB$

$$A \rightarrow a$$

$$B \rightarrow b$$

$\Rightarrow S \rightarrow AB$

B comes immediate after A

$$\therefore \text{FOLLOW}(A) = \text{FIRST}(B)$$

We know

$$\text{FIRST}(B) = \{b\}$$

$$\text{FOLLOW}(A) = \{b\}$$

Now B is at the end

$$\therefore \text{FOLLOW}(B) = \{\$ \}$$

Eg: ⑤ $S \rightarrow Abc$

$$\text{FOLLOW}(S) = \{\$ \}$$

$$\text{FOLLOW}(A) = \{b\}$$

Eg (6) $S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b | \epsilon$

$FOLLOW(A) = FIRST(B)$

$FIRST(B) = \{b, \epsilon\}$

FOR ~~ε~~ FOLLOW, we do not put ε.

$S \rightarrow A \overset{\epsilon}{\cancel{B}}$

∴ A can effectively become the last symbol.

$FOLLOW(S) \subseteq FOLLOW(A)$

$FOLLOW(A) = \{b, \$ \}$

Eg (7) $S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow \epsilon$

$FOLLOW(A) = FIRST(B)$

$FIRST(B) = \{\epsilon\}$

$S \rightarrow A \overset{\epsilon}{\cancel{B}}$

$FOLLOW(A) = \{\$ \}$

\$ is only for start symbol which is E

Eg ⑧ $S \rightarrow a S b \mid \epsilon$

$$\text{FOLLOW}(S) = \{ \$, b \}$$

Eg ⑨ $S \rightarrow a S b \mid \epsilon$

$$A \rightarrow S +$$

$$\text{FOLLOW}(S) = \{ \$, b, + \}$$

Q. Consider the grammar:

$$E \rightarrow T E_1, E_1 \rightarrow + T E_1 \mid \epsilon$$

$$T \rightarrow F T_1, T_1 \rightarrow * F T_1 \mid \epsilon$$

$$F \rightarrow (E) \mid \text{id}$$

Find the FOLLOW set of all the variables.

→ All the variables:

$$E, E_1, T, T_1, F$$

$$\boxed{\text{FOLLOW}(E) = \{ \$,) \}}$$

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 FOLLOW(E_1) = ?

~~$E \rightarrow TE_1$~~
 ~~$FOLLOW(E_1) = FOLLOW(E)$~~
 ~~$= \{ \$,) \}$~~
and

$E \rightarrow TE_1$

E_1 is at the end of production

$\therefore FOLLOW(E) \subseteq FOLLOW(E_1)$

$FOLLOW(E_1) = \{ \$,) \}$

FOLLOW(T) = ?

~~$E \rightarrow TE_1$~~
 ~~$= \{ \$,) \}$~~

and,

$E_1 \rightarrow +TE_1 \mid \epsilon$

$FOLLOW(T) = FIRST(E_1)$

$\rightarrow$ this we have found before.

$FIRST(E_1) = \{ +, \epsilon \}$

~~Follow(T) = { \$, >, + }~~

$$E_1 \xrightarrow{\epsilon} +T \epsilon_1$$

~~Follow(T)~~ $\text{Follow}(E_1) \subseteq \text{Follow}(T)$

$$\boxed{\text{Follow}(T) = \{ \$, >, + \}}$$

$$\text{Follow}(T_1) = ?$$

~~$T_1 \xrightarrow{\epsilon} E_1 \epsilon$~~
 $= \$$

~~and,~~

$$T \rightarrow FT_1$$

$$\text{Follow}(T) \subseteq \text{Follow}(T_1)$$

$$\boxed{\text{Follow}(T_1) = \{ \$, >, + \}}$$

$$\text{FOLLOW}(F) = \{$$

~~$$\text{FOLLOW}(F) = \{$$~~

$$= \{ \$ \}$$

~~$\{ \$ \}$~~

$$T \rightarrow FT_1$$

$$\text{FOLLOW}(F) = \text{FIRST}(T_1)$$

$$\text{FIRST}(T_1) = \{ *, \epsilon \}$$

$$T \rightarrow F \overset{\epsilon}{T_1}$$

$$\text{FOLLOW}(F) = \text{FOLLOW}(T)$$

and,

$$T_1 \rightarrow *FT_1 \mid \epsilon$$

$$\text{FOLLOW}(F) = \text{FIRST}(T_1) \text{ same result}$$

$$\boxed{\text{FOLLOW}(F) = \{ \$, *, >, + \}}$$

\$ only for Start symbol which is S.

Q. Find the FOLLOW from the following production:

$S \rightarrow a B D h$, $B \rightarrow c C$, $C \rightarrow b C | e$, $D \rightarrow E F$, $E \rightarrow g | e$,
 $F \rightarrow f | e$.

all the variable.

S, B, C, D, E, F

$$\text{FOLLOW}(S) = \{ \$ \}$$

$$\text{FOLLOW}(B) = \{ \$, \text{FIRST}(D) \}$$

$$\text{FOLLOW}(B) = \{ \$, g, f, h \}$$

$$\text{FOLLOW}(C) = \{ \$, g, f, h \}$$

$$\text{FOLLOW}(D) = \{ \$, h \}$$

$$\text{FOLLOW}(E) = \{ \$, f, h \}$$

$$\text{FOLLOW}(F) = \{ \$, h \}$$